SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 30%

QUESTIONS: 3 **Total Marks:30**

Annexure C

Case study

Criteria	Understanding of Case ***(2.5)***	Application of Concepts ***(2.5)***	Analysis ***(2)***	Solution Design ***(2)***	Clarity ***(1)***	***Total(10)***
Weightage	*25%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						

Code of Conduct:

I Hemalakshmi H (192571049) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANNEXURE A

Case Study:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP,FP,FN,TN

2.Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP,FP,FN,TN and Calculate the Precision and Recall and F1 Score .

3.An email service provider wants to classify emails as **Spam or Not Spam** using features:

- Contains “Offer” (Yes/No)
- Contains “Free” (Yes/No)
- Email Length (Short/Long)

Sample Dataset

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:
    - P(Spam), P(Not Spam)
2. Compute conditional probabilities for each feature.
3. Using **Naïve Bayes**, classify:
    - (Offer=Yes, Free=No, Length=Short)
4. Compare results using **WEKA (Naïve Bayes classifier)**.
5. Discuss the **independence assumption** in Naïve Bayes

Q1:

Confusion Matrix Interpretation for “The Boy Who Cried Wolf”

In this case study, the shepherd boy acts as a predictor of danger. His shout “Wolf!” represents a warning signal, and the actual presence or absence of a wolf represents the true condition. The confusion matrix helps classify each situation based on whether the boy’s warning matches reality.

True Positive (TP)

A True Positive occurs when the boy correctly identifies the presence of a wolf. In the story, this happens during the night when the real wolf approaches the flock. The boy sees the wolf and immediately shouts “Wolf!” This action is a correct prediction because the wolf is truly present. Even though the villagers do not respond, the boy’s detection itself is accurate. Therefore, this moment represents a True Positive, where the predicted danger matches the actual danger.

False Positive (FP)

A False Positive occurs when the boy predicts a wolf even though no wolf is present. This corresponds to the two earlier incidents where the boy becomes bored and shouts “Wolf!” purely for amusement. The villagers rush to protect the flock, only to discover that the boy was lying. These prank calls are classic examples of False Positives because the boy signals danger when there is none. The villagers’ wasted effort and frustration highlight the consequences of repeated false alarms.

False Negative (FN)

A False Negative occurs when a wolf is present, but the warning is not acted upon or is effectively ignored. In the story, this happens when the real wolf appears and the boy again shouts “Wolf!” However, due to his previous false alarms, the villagers assume he is lying and choose not to respond. As a result, the wolf attacks the flock. Although the boy did shout, the system (the villagers) behaves as if there is no wolf. This situation represents a False Negative because the actual danger is present, but the response fails to acknowledge it.

True Negative (TN)

A True Negative occurs when no wolf is present and no warning is given. This represents the normal, uneventful periods when the boy quietly tends the flock without shouting “Wolf!” and there is no wolf nearby. In these moments, the absence of a warning correctly matches the absence of danger. This is a True Negative because both the prediction and the actual condition indicate that everything is safe.

Q2:

Confusion Matrix Identification

You are given:

- Total images = 100
- Cat images = 30
- No-cat images = 70

Model performance:

- Correctly predicts cat in **25 of 30** cat images → True Positives
- Correctly predicts no-cat in **50 of 70** no-cat images → True Negatives

From this, we can derive all four confusion-matrix values.

True Positive (TP)

Model predicts **cat**, and the image **actually contains a cat**. TP = 25

False Negative (FN)

Model predicts **no cat**, but the image **actually contains a cat**. Total cat images = 30 TP = 25 FN = 30 − 25 = 5

True Negative (TN)

Model predicts **no cat**, and the image **actually does not contain a cat**. TN = 50

False Positive (FP)

Model predicts **cat**, but the image **actually does not contain a cat**. Total no-cat images = 70
TN = 50 FP = 70 − 50 = 20

Final Confusion Matrix Values

- **TP = 25**
- **FP = 20**
- **FN = 5**
- **TN = 50**

Metric Calculations

1. Precision

$$Precision = \frac{TP}{TP + FP}$$

$$Precision = \frac{25}{25 + 20} = \frac{25}{45}$$

$$Precision = 0.5556 \approx 0.56$$

2. Recall

$$Recall = \frac{TP}{TP + FN}$$

$$Recall = \frac{25}{25 + 5} = \frac{25}{30}$$

$$Recall = 0.8333 \approx 0.83$$

3. F1 Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Substitute values:

$$F1 = 2 \times \frac{0.5556 \times 0.8333}{0.5556 + 0.8333}$$

$$F1 = 2 \times \frac{0.463}{1.3889}$$

$$F1 = 2 \times 0.3333 = 0.6666$$

$$F1 \approx 0.67$$

Final Answers

- **TP = 25**
- **FP = 20**
- **FN = 5**
- **TN = 50**
- **Precision ≈ 0.56**
- **Recall ≈ 0.83**
- **F1 Score ≈ 0.67**

Q3:

Given Dataset

Offer	**Free**	**Length**	**Class**
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Total samples = 6 Spam = 3 Not Spam = 3

1. Prior Probabilities

$$P(Spam) = \frac{3}{6} = 0.5$$

$$P(Not\ Spam) = \frac{3}{6} = 0.5$$

2. Conditional Probabilities

We compute probabilities **feature-wise** for each class.

For Class = Spam

Spam rows:

1. Yes, Yes, Short
2. No, Yes, Short
3. Yes, Yes, Long

Total Spam = 3

Offer

- Offer = Yes → 2 out of 3
- Offer = No → 1 out of 3

$$P(Offer = Yes \mid Spam) = \frac{2}{3}$$

$$P(Offer = No \mid Spam) = \frac{1}{3}$$

Free

- Free = Yes → 3 out of 3
- Free = No → 0 out of 3

$$P(Free = Yes \mid Spam) = 1$$

$$P(Free = No \mid Spam) = 0$$

Length

- Short → 2 out of 3
- Long → 1 out of 3

$$P(Length = Short \mid Spam) = \frac{2}{3}$$

$$P(Length = Long \mid Spam) = \frac{1}{3}$$

For Class = Not Spam

Not-Spam rows:

1. Yes, No, Long
2. No, No, Long
3. No, No, Short

Total Not Spam = 3

Offer

- Offer = Yes → 1 out of 3
- Offer = No → 2 out of 3

$$P(Offer = Yes \mid NotSpam) = \frac{1}{3}$$

$$P(Offer = No \mid NotSpam) = \frac{2}{3}$$

Free

- Free = Yes → 0 out of 3
- Free = No → 3 out of 3

$$P(Free = Yes \mid NotSpam) = 0$$

$$P(Free = No \mid NotSpam) = 1$$

Length

- Short → 1 out of 3
- Long → 2 out of 3

$$P(Length = Short \mid NotSpam) = \frac{1}{3}$$

$$P(Length = Long \mid NotSpam) = \frac{2}{3}$$

3. Classify Using Naïve Bayes

Input to classify:

Offer = Yes Free = No Length = Short

We compute:

$$P(Spam \mid X) \propto P(Spam) \times P(Offer = Yes \mid Spam) \times P(Free = No \mid Spam) \times P(Length = Short \mid Spam)$$

$$P(NotSpam \mid X) \propto P(NotSpam) \times P(Offer = Yes \mid NotSpam) \times P(Free = No \mid NotSpam) \times P(Length = Short \mid NotSpam)$$

Compute Spam probability

$$P(Spam) = 0.5$$

$$P(Offer = Yes \mid Spam) = \frac{2}{3}$$

$$P(Free = No \mid Spam) = 0$$

$$P(Length = Short \mid Spam) = \frac{2}{3}$$

$$P(Spam \mid X) = 0.5 \times \frac{2}{3} \times 0 \times \frac{2}{3} = 0$$

Because one feature probability is zero, the entire product becomes zero.

Compute Not-Spam probability

$$P(NotSpam) = 0.5$$

$$P(Offer = Yes \mid NotSpam) = \frac{1}{3}$$

$$P(Free = No \mid NotSpam) = 1$$

$$P(Length = Short \mid NotSpam) = \frac{1}{3}$$

$$P(NotSpam \mid X) = 0.5 \times \frac{1}{3} \times 1 \times \frac{1}{3}$$

$$P(NotSpam \mid X) = 0.5 \times \frac{1}{9}$$

$$P(NotSpam \mid X) = \frac{1}{18} \approx 0.0556$$

Final Classification

Since:

- $P(Spam \mid X) = 0$
- $P(NotSpam \mid X) > 0$

The email is classified as: Not Spam

4. Compare with WEKA (Naïve Bayes)

If you load this dataset into WEKA:

- WEKA also uses Naïve Bayes
- It will compute the same conditional probabilities
- Because $P(Free = No \mid Spam) = 0$, WEKA will also classify the email as **Not Spam**

(WEKA uses Laplace smoothing only if enabled; default Naïve Bayes in WEKA does not smooth zero probabilities.)

Thus, WEKA will give the same result: **Not Spam**.

5. Independence Assumption in Naïve Bayes

Naïve Bayes assumes that all features are **independent** of each other given the class.

In this email example, it assumes:

- Whether the email contains “Offer” does not depend on whether it contains “Free”
- Email length does not depend on the presence of “Offer” or “Free”

In reality, this assumption is often false. For example:

- Emails containing “Offer” often also contain “Free”
- Spam emails often have similar structure and length

Even though this independence assumption is unrealistic, Naïve Bayes still performs well in many text-classification tasks because the probability ratios still separate classes effectively.